Section 6

Micro – CC411

- ADD vs ADC

ADD Ax, Bx **Ax = Ax + Bx**	**ADC Ax, Bx** **Ax = Ax + Bx + C.F.**

- CLC: C.F. = 0
- STC: C.F. = 1

Example

Mov Ax,4
Add Ax,5 <- Ax=4+5
- - - -

Question 1

What are the values of Ax, CF, AF, and ZF after the following:

```
Mov Ax,4
CLC
Adc Ax, 5
```

Question 1

What are the values of CF, AF, and ZF after the following:

Mov Ax,4

CLC

Adc Ax, 5 <- Ax=4+5+C.F(0).

00000100

+

00000101

+

0 <- C.F.

00001001 C.F. = 0//A.F.=0 //Z.F.=0

Assembly

Structure of An Assembly program

```
.model small
.stack 128
.data

.code
main proc far

    MOV AX,@DATA
    MOV DS,AX



    mov ax, 4c00h ; exit to operating system.
    int 21h

main endp
end main ; set entry point and stop the assembler.
```

Size of Program

Size of Stack

Data

Code

Simple Program

Question 2

Write, run, and analyze a program that calculates the total sum paid to a salesperson for eight months. The following are the monthly paychecks for those months: $2300, $4300, $1200, $3700, $1298, $4323, $5673, $986.

Question 2

Cx=8 **Bx=0**

2300	4300	1200	3700	1298	4323	5673	986

SI

adc bx,[si]

Cx=7

Question 2

Bx=2300

2300	4300	1200	3700	1298	4323	5673	986

SI

adc bx,[si]

Cx=6

Question 2

Bx=6600

2300	4300	1200	3700	1298	4323	5673	986

SI

adc bx,[si]

Cx=5

Question 2

Bx=7800

2300	4300	1200	3700	1298	4323	5673	986

SI

adc bx,[si]

Cx=4

Question 2

Bx=11500

2300	4300	1200	3700	1298	4323	5673	986

SI

adc bx,[si]

Cx=3

Question 2

Bx=12798

2300	4300	1200	3700	1298	4323	5673	986

SI

Cx=2

adc bx,[si]

Question 2

Bx=17212

2300	4300	1200	3700	1298	4323	5673	986

SI

Cx=1

adc bx,[si]

Question 2

Bx=22794

2300	4300	1200	3700	1298	4323	5673	986

SI

Cx=0

adc bx,[si]

mov sum,bx

Question 2

Bx=23780

2300	4300	1200	3700	1298	4323	5673	986

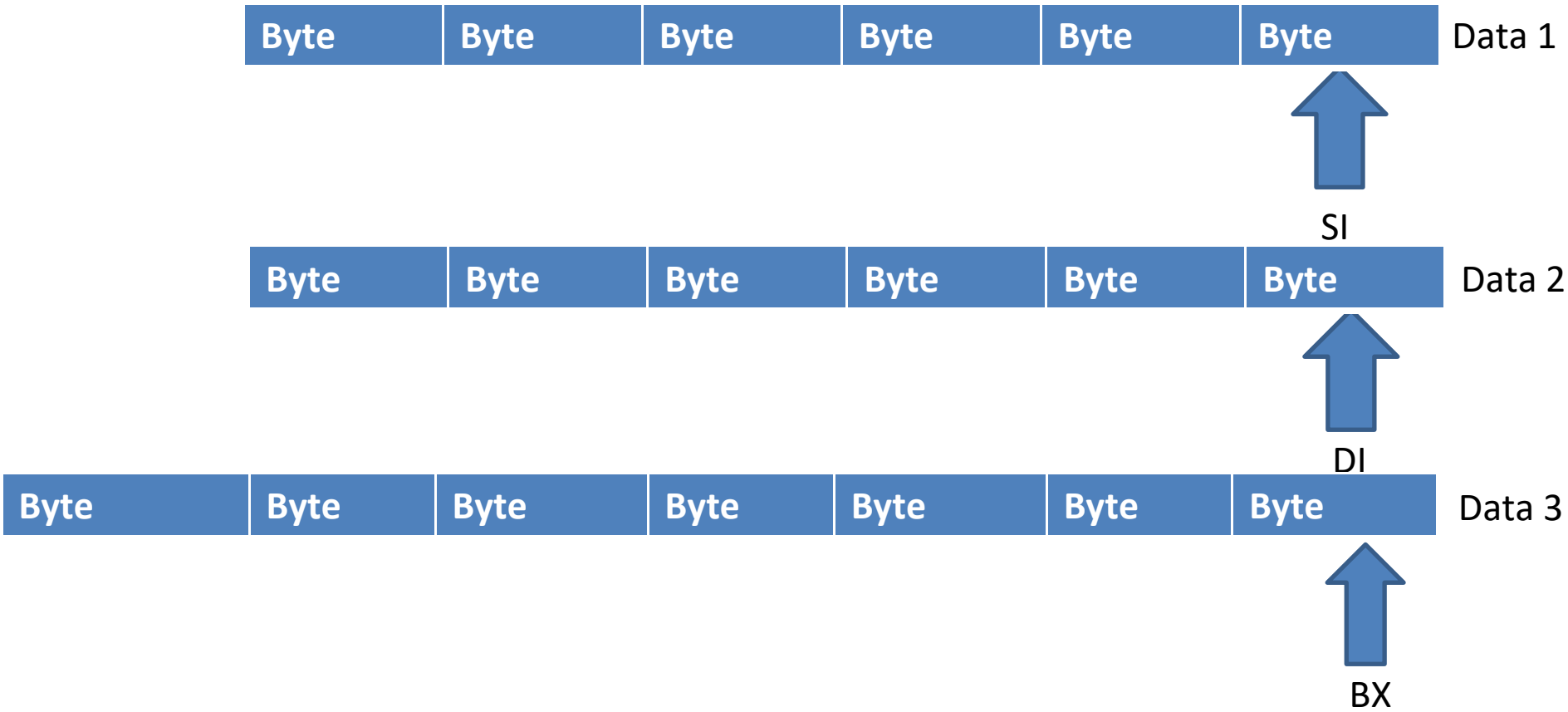
Cx=0

OUT OF LooP

Question 3

Perform Byte addition on two 6 bytes numbers.

Question 3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Data 2
DI
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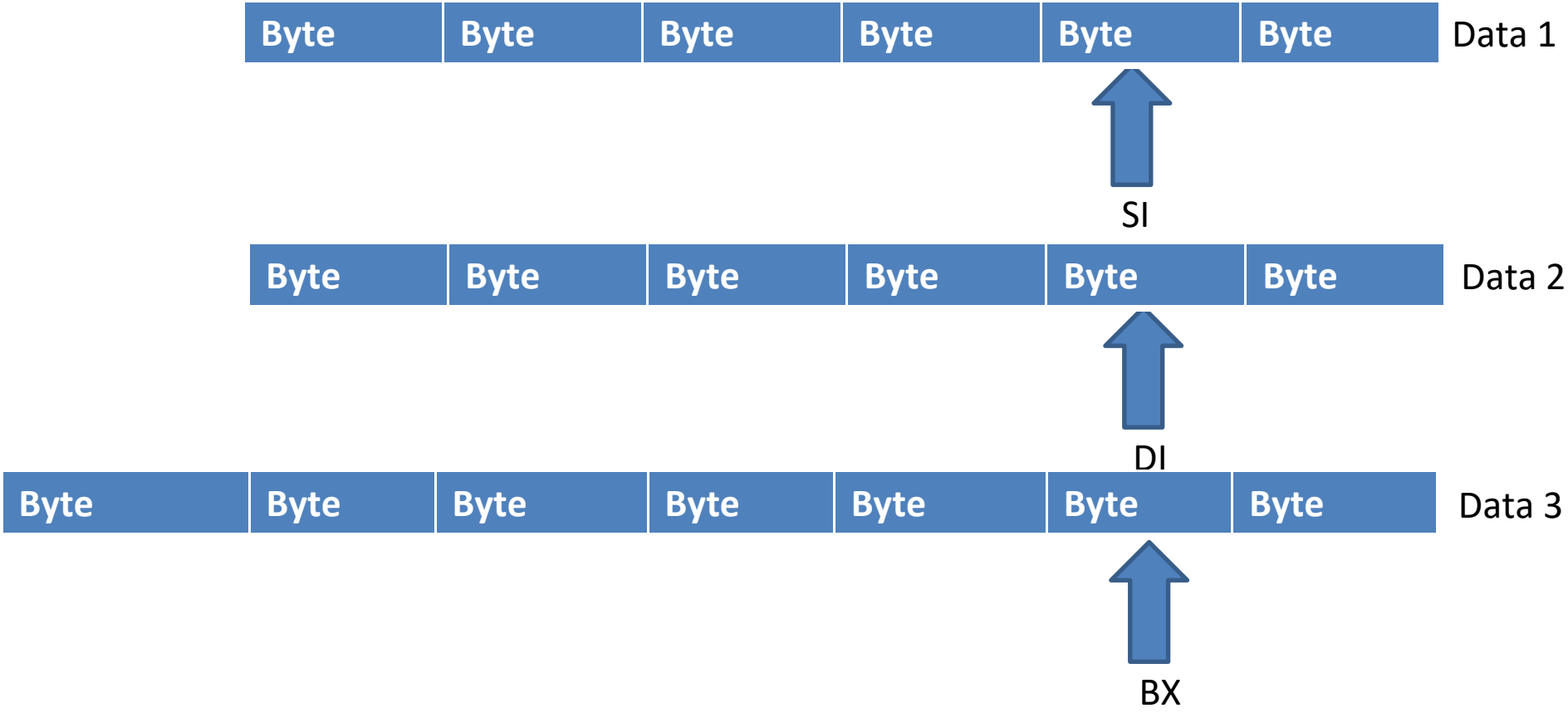
Byte
Byte
Byte
Data 3
BX

Question 3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Data 3
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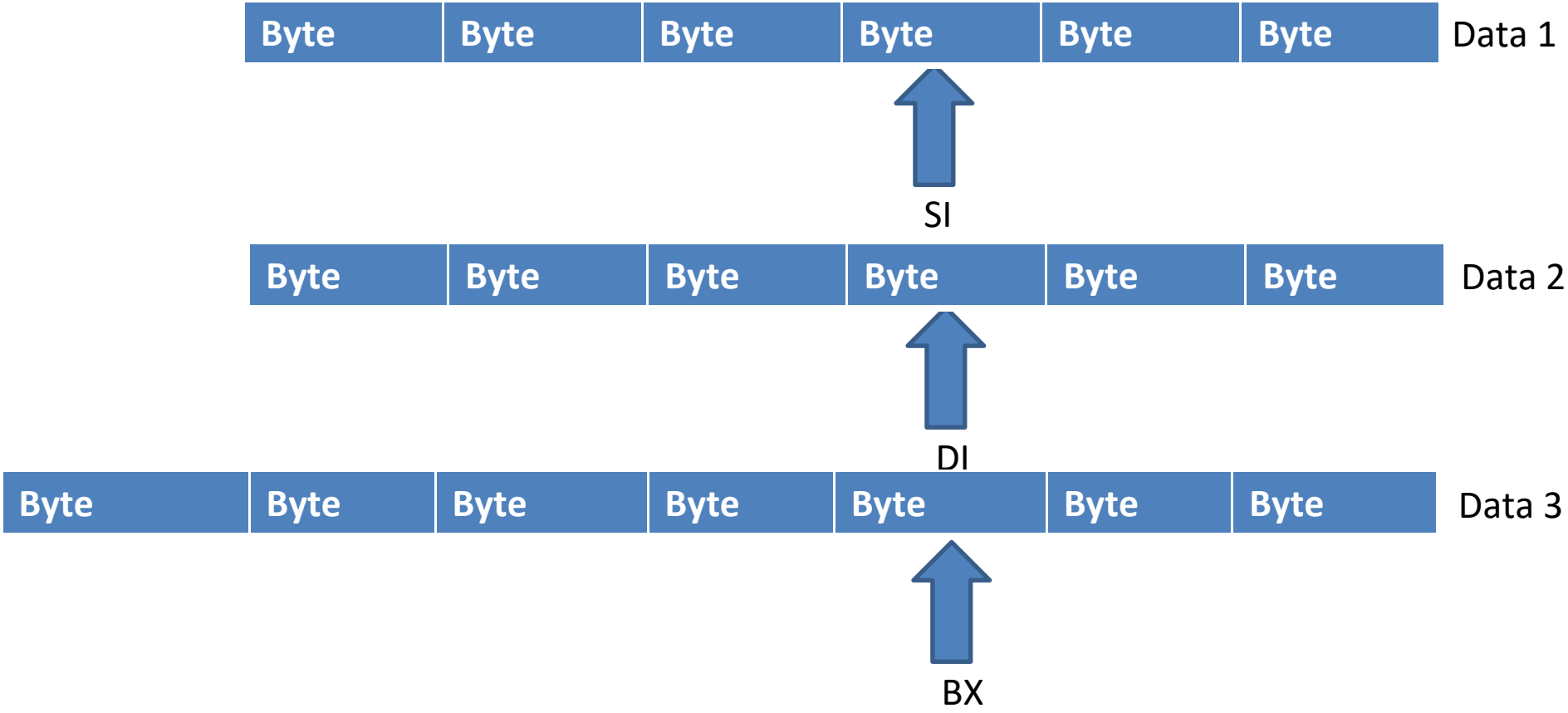
Question 3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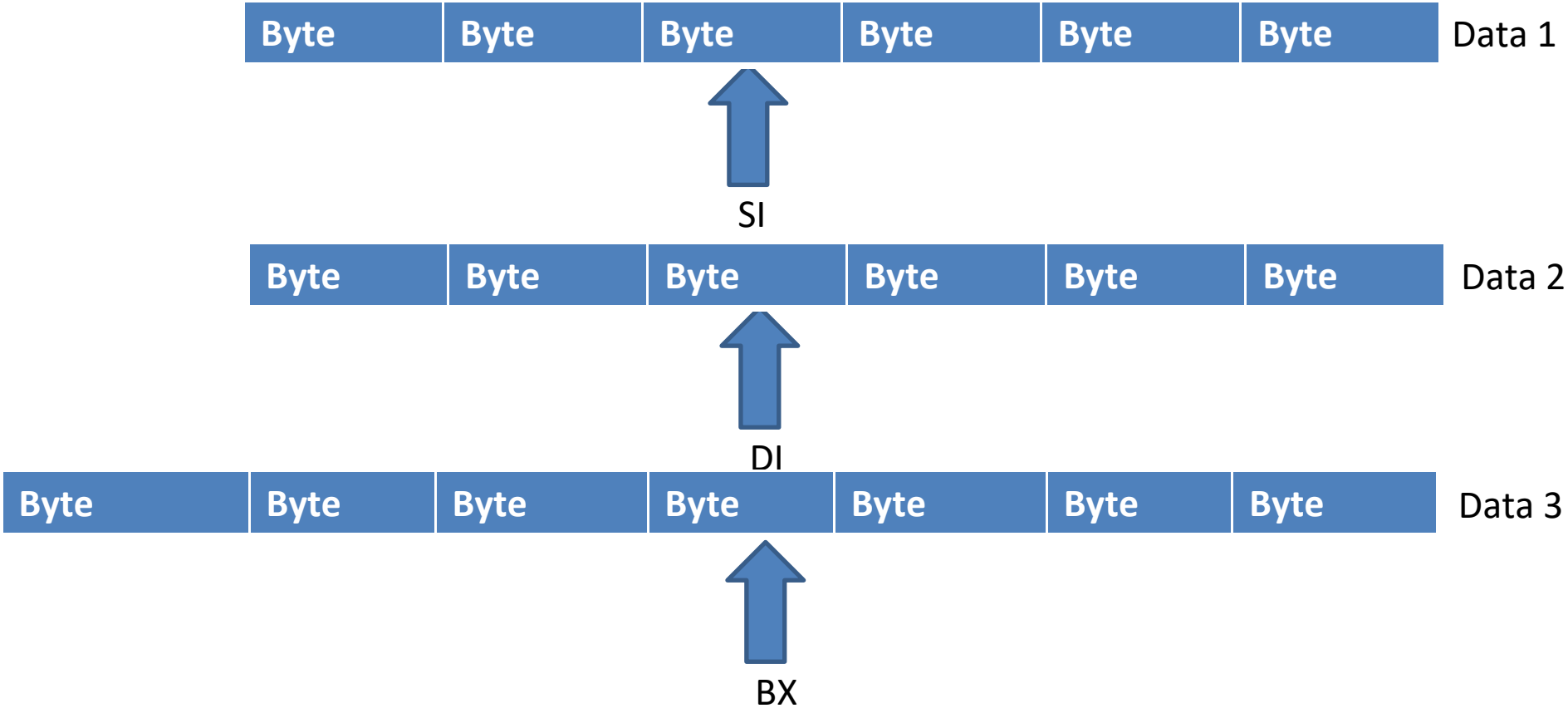
Data 3
BX

Question 3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Data 3
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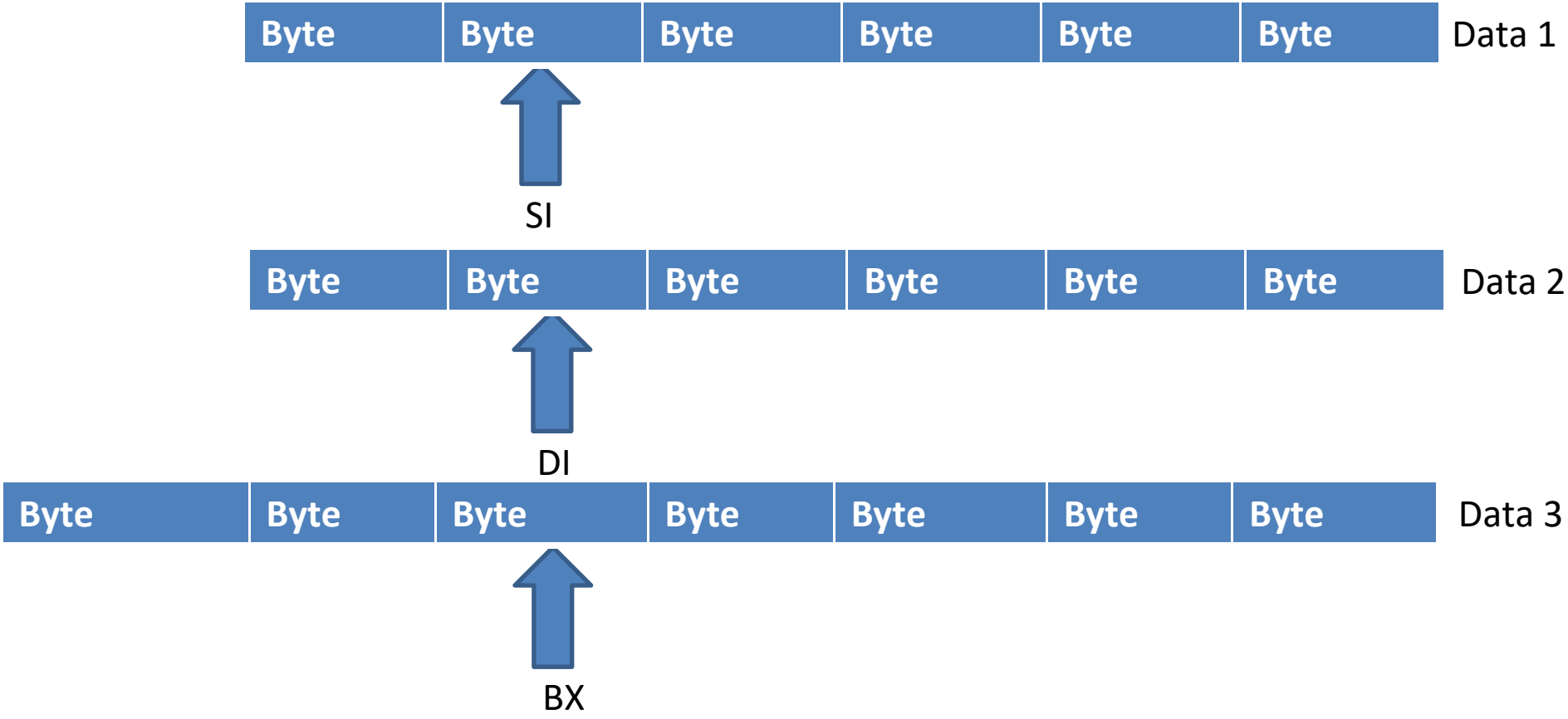
Question 3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Data 3
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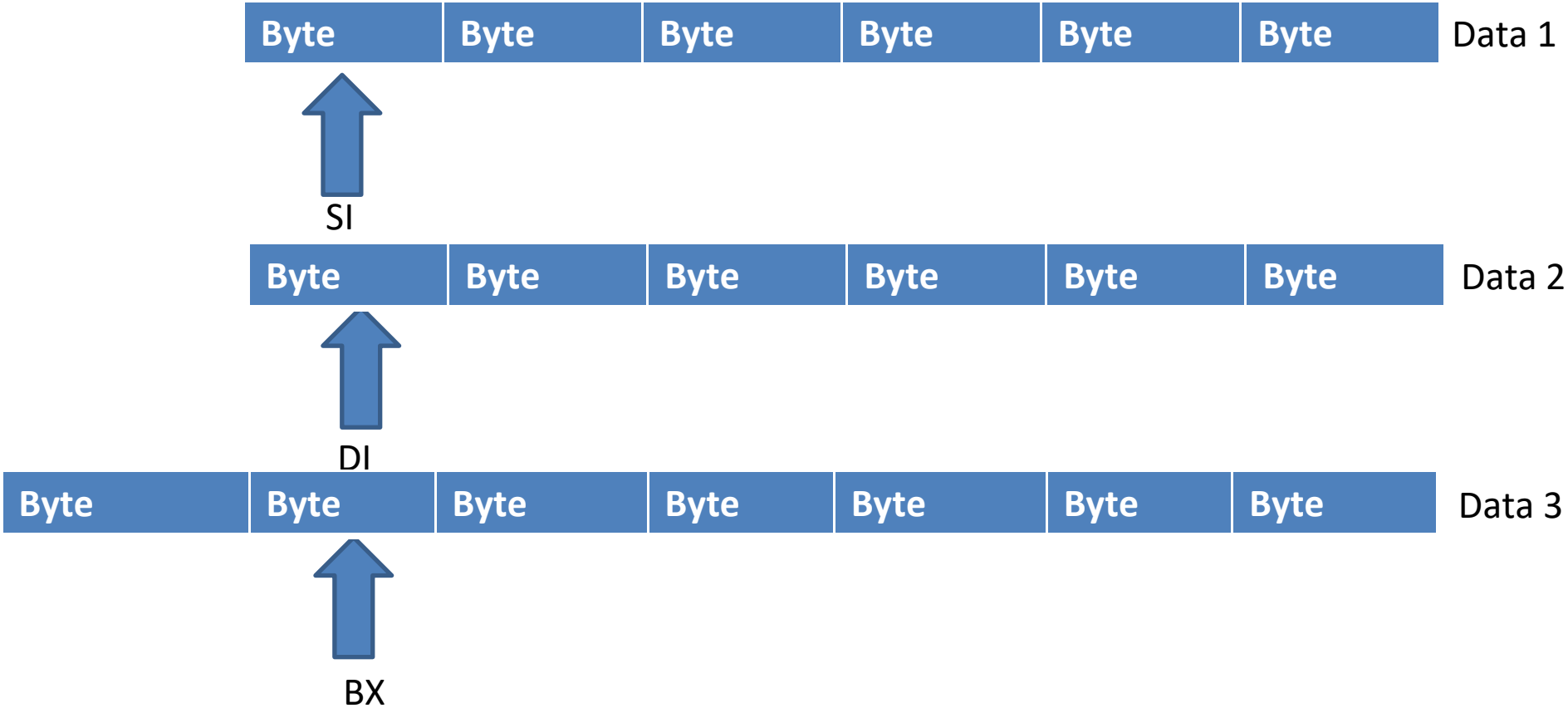
Question 3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Byte
Data 3
BX

- SUB vs SBB

SUB Ax,Bx **Ax=Ax-Bx**	**SBB Ax,Bx** **Ax=Ax-Bx-cf**

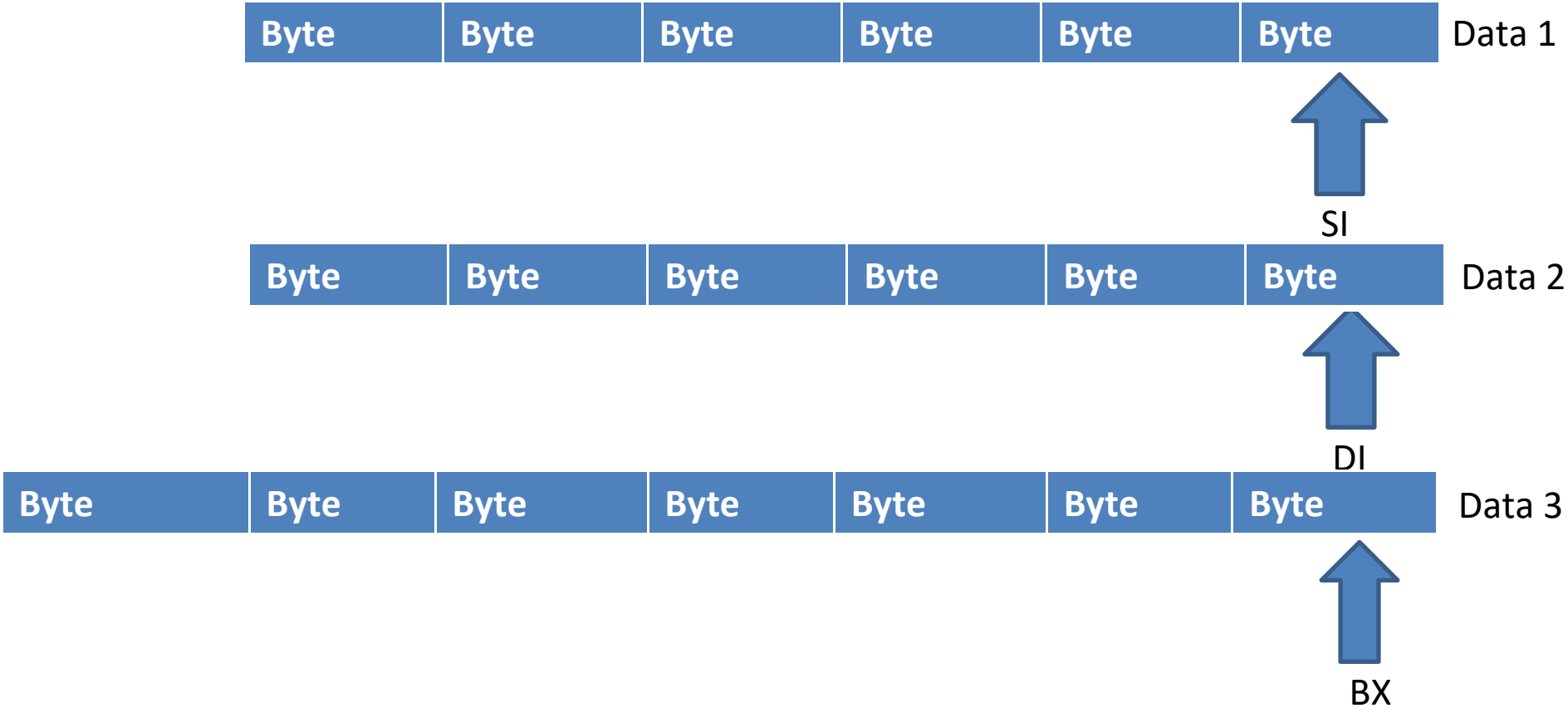
iF CF is 0 SBB is SUB

Else SBB is A - B - 1

Question 4

Perform Byte Subtraction on two 6 bytes numbers.

Question 4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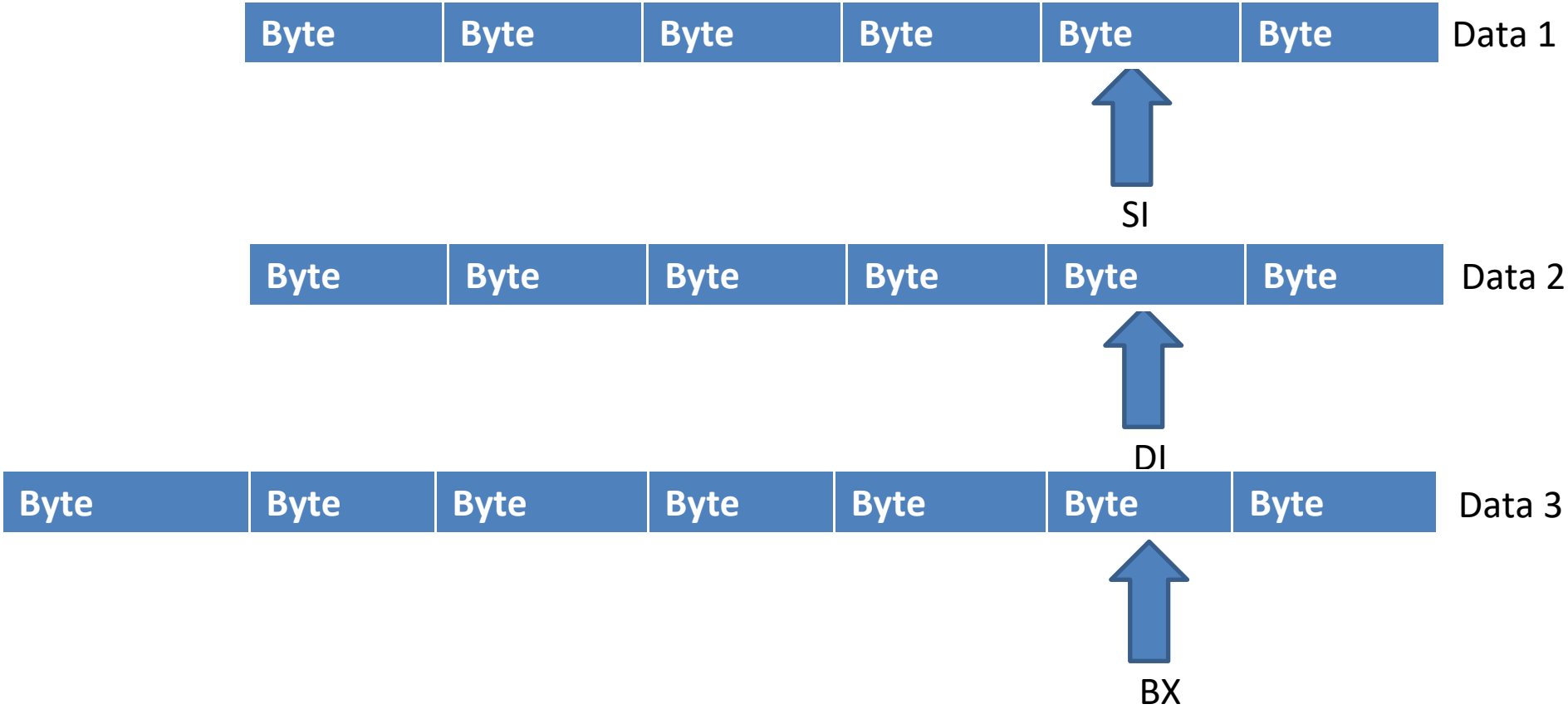
Data 3
BX

Question 4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Data 2
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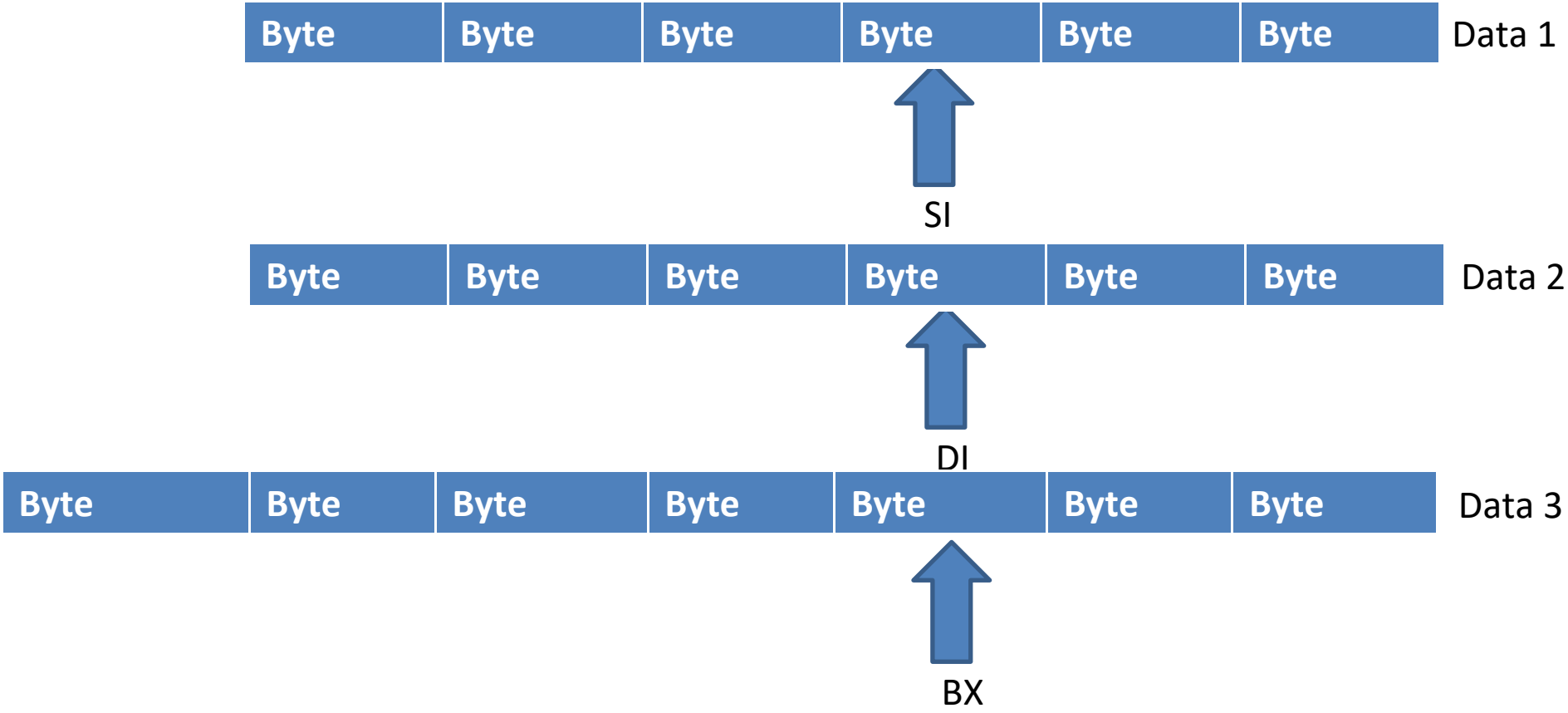
Byte
Data 3
BX

Question 4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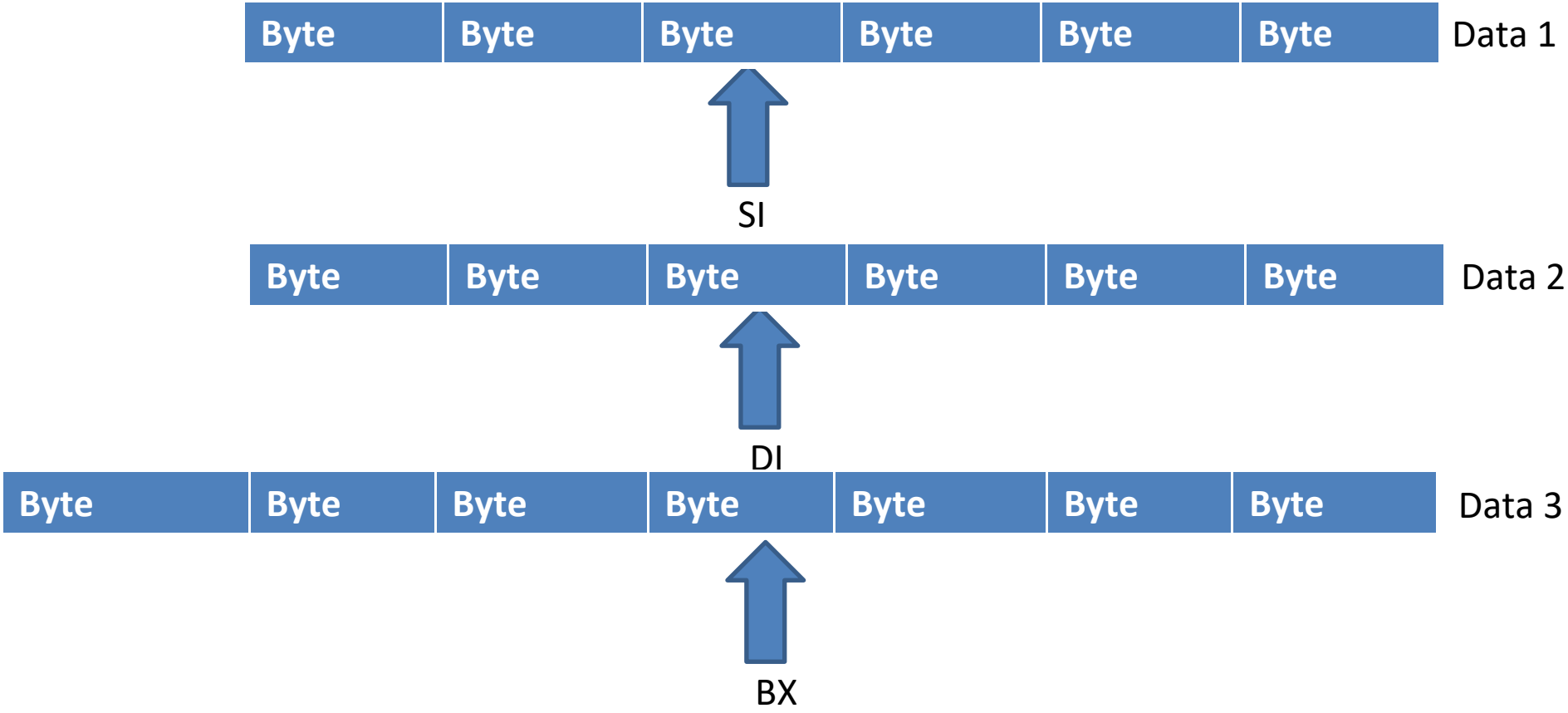
Data 3
BX

Question 4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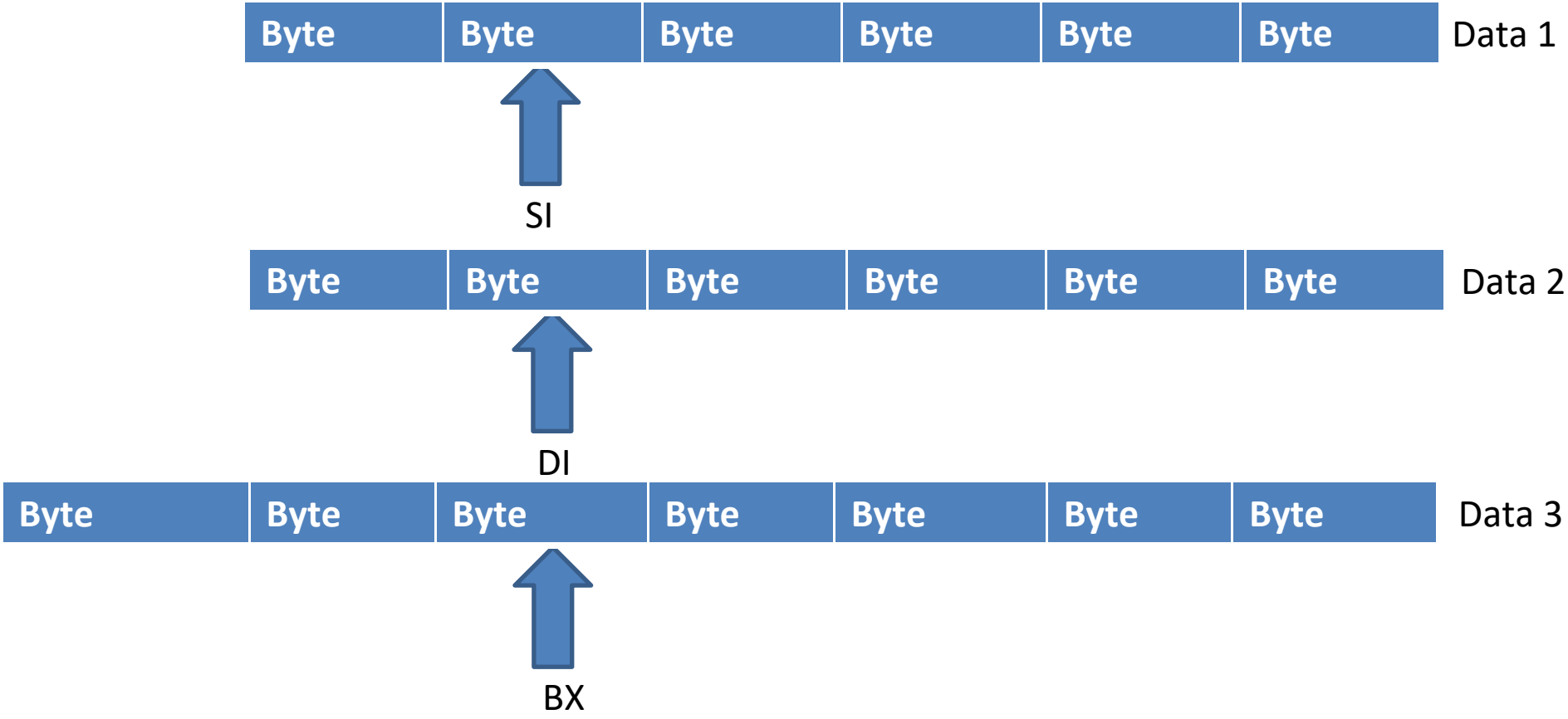
Byte
Data 3
BX

Question 4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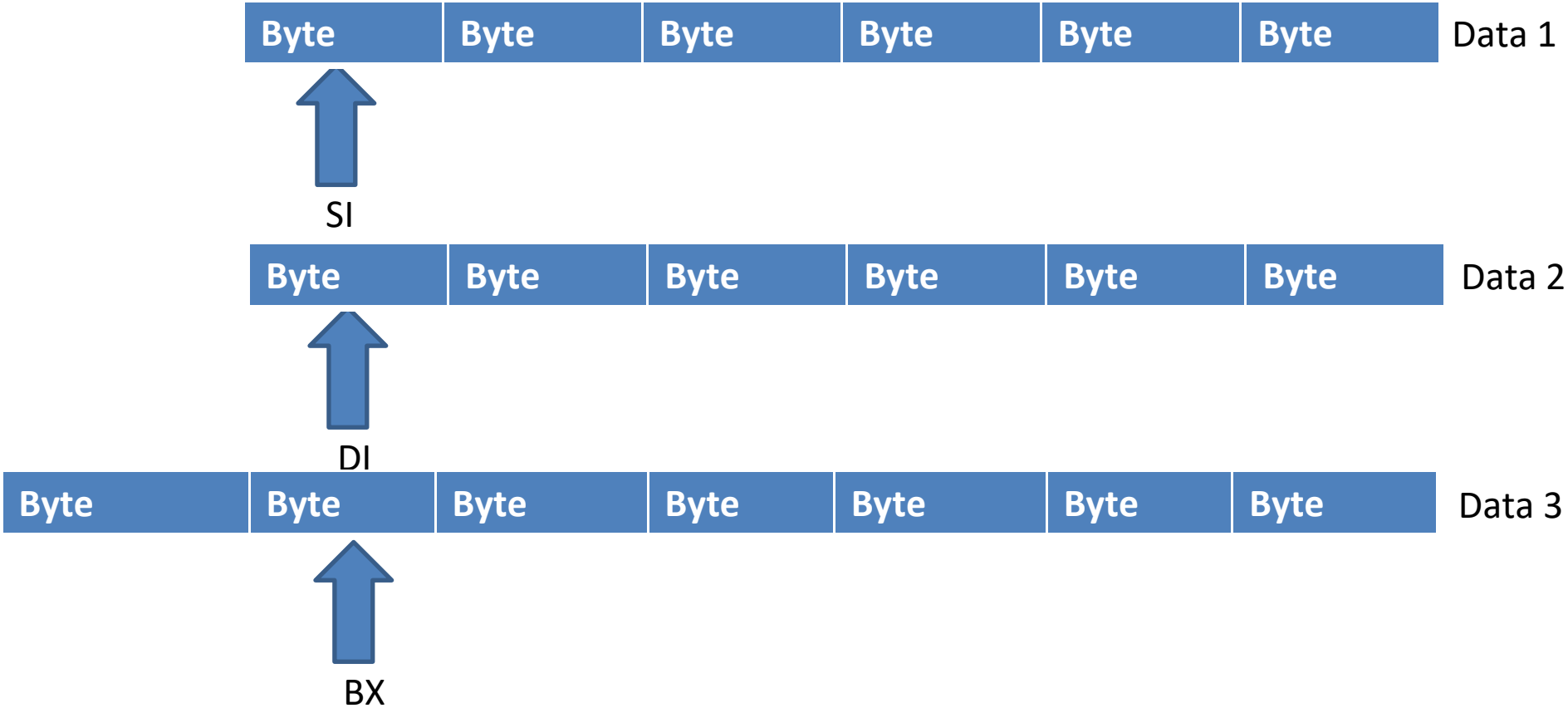
Data 3
BX

Question 4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Byte
Byte
Data 3
BX

Question 5

State the three steps involved in a SUB and show the steps for the following data.

a) 23H – 12H

23H + (-12H)

Step 1: 2's complement of 2nd number

Step 2: ADD

Step 3: invert CF

Question 5

23H – 12H

0010 0011 – 0001 0010

1) 2's Comp. 0001 0010

1110 1101 +1

1110 1110 ←

2) ADD 00100011

11101110

(1)00010001

3) Invert Carry Flag: C.F. = 0

Multiplication

Table 3-1: Unsigned Multiplication Summary

Multiplication	Operand 1	Operand 2	Result
byte × byte	AL	register or memory	AX
word × word	AX	register or memory	DX AX
word × byte	AL = byte, AH = 0	register or memory	DX AX

Table 3-1: Unsigned Multiplication Summary

Multiplication	Operand 1	Operand 2	Result
byte × byte	AL	register or memory	AX
word × word	AX	register or memory	DX AX
word × byte	AL = byte, AH = 0	register or memory	DX AX

Assume byte1 = 230, byte2 = 100, word1 = 9998, word2 = 300, and doubleword = 100000.

(a) byte1 × byte2

```
Mov AL, 230
Mov  BL,100
MUL BL
```

(b) byte1 × word1

```
Mov AL, 230
Mov Ah,0
Mov BX,9998
MUL BX
```

(c) word1 × word2

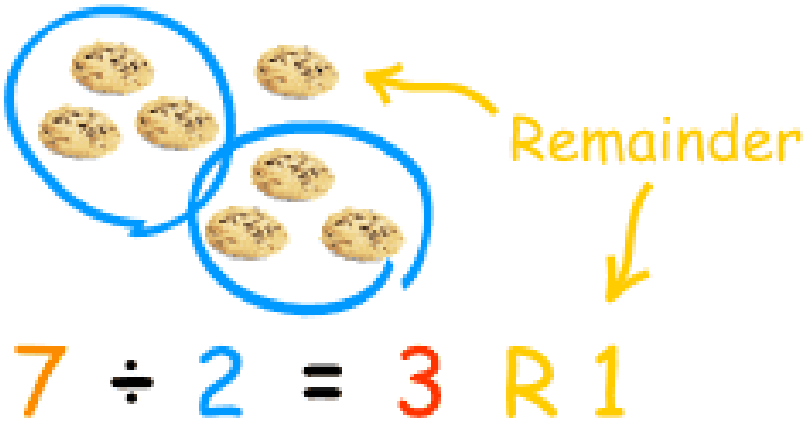
```
Mov Ax, 9998
Mov Bx, 300
MUL BX
```

Division

Table 3-2: Unsigned Division Summary

Division	Numerator	Denominator	Quotient	Rem
byte/byte	AL = byte, AH = 0	register or memory	AL [1]	AH
word/word	AX = word, DX = 0	register or memory	AX [2]	DX
word/byte	AX = word	register or memory	AL [1]	AH
doubleword/word	DXAX = doubleword	register or memory	AX [2]	DX

Remainder



7/2 = 3.5

0.5*2= $\boxed{1}$

Table 3-2: Unsigned Division Summary

Division	Numerator	Denominator	Quotient	Rem
byte/byte	AL = byte, AH = 0	register or memory	AL [1]	AH
word/word	AX = word, DX = 0	register or memory	AX [2]	DX
word/byte	AX = word	register or memory	AL [1]	AH
doubleword/word	DXAX = doubleword	register or memory	AX [2]	DX

Assume byte1 = 230, byte2 = 100, word1 = 9998, word2 = 300, and doubleword = 100000 (186A0H).

(a) byte1 / byte2

```
Mov AL, 230
Mov Ah,0
Mov BL,100
Div BL
```

(b) word1 / word2

```
Mov Ax, 9998
Mov Dx,0
Mov Bx,300
Div BX
```

(c) doubleword / byte1

```
Mov Ax,86A0H
Mov Dx,0001H
Mov Bx,100
Div Bx
```